

DANIELE PESSINA

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EDUCATION

Imperial College London

2022 – Present

PhD in Chemical Engineering - CASE Studentship with AstraZeneca

- Thesis title: A hybrid *in-silico/in-vitro* approach for integrated design and optimisation of large-molecule crystallisation

Imperial College London

2018 – 2022

MEng in Chemical Engineering, 1st Class Honours

EXPERIENCE

Modeling & Optimization Scientist

May 2026 - Present

Deep Origin, Cell Sims team - ARPA-H CATALYST program (PREDICTS consortium)

- Built a differentiable PyTorch framework that links mechanistic models of human physiology with machine-learning components into a single trainable system that predicts patient-level drug toxicity.
- Integrated heterogeneous simulation engines (C++ and Julia solvers, SBML models, agent-based simulators) into a single differentiable platform, with trainable ML leaves interwoven between the mechanistic stages.
- Calibrated the framework's predictions against clinical data and built interpretability tools to rank the main parameter and transcriptomic drivers of toxicity.

PhD Researcher, Process Systems Engineering

2022 - Present

Supervision: Dr. Maria Papathanasiou & Prof. Jerry Y. Y. Heng

- Developed and deployed physics-informed novel ML architectures (neural ODEs, transformers, autoencoders), coupled with tools such as transfer learning, reinforcement learning and hybrid modelling for improved time-series modelling of sparse and multi-fidelity datasets of biological dynamical systems. Applied neuro-symbolic methodologies for enhanced interpretability and reduced model complexity in crystallisation systems.
- Explored uncertainty quantification methods for PDE- and ML-based models. Built autodifferentiable probabilistic inference pipelines in JAX and Julia, which enabled uncertainty-aware parameter estimation, robust system optimisation and process controllability assessments.
- Built hybrid pharmacokinetic models combining mechanistic PK equations with neural networks trained on multimodal clinical data (drug administration records, electroencephalogram signals and covariates) for personalised Remifentanyl modelling and patient stratification. Used physics-informed Neural ODEs and autoregressive components to capture drug dynamics and flag anomalous trajectories under distribution shift.
- Designed ML-enhanced sensitivity analysis, model reduction and explainability tools for complex and intractable crystallisation and pharmacokinetic models.
- Conducted *in-vitro* medium-scale lysozyme crystallisation experiments with MT EasyMax and PAT tools, generating experimental evidence for template-induced protein nucleation mechanisms.

Data Scientist Intern

Jan. - Apr. 2026

Quaisr, London

- Collaborated directly with pharmaceutical clients to architect and deploy agentic AI workflows for regulated industries, combining LLM-driven automation with domain-specific data science for end-to-end process development, from experimental design through production deployment.

Consultant Data Scientist

March - July 2025

Haleon

- Delivered ML-based soft sensors providing real-time product quality estimation from high-frequency manufacturing data, reducing reliance on offline batch testing.
- Benchmarked LSTM and transformer models against classical feature-engineered approaches, improving adaptability and robustness under imbalanced datasets and data augmentation strategies.

TECHNICAL SKILLS

Core: SciML, probabilistic programming, ODE/PDE numerical modelling, Bayesian inference and optimisation, uncertainty quantification, differentiable & multiscale modelling, PBPK/QSP simulation, interpretability (SHAP, PDP/ICE), time-series modelling, surrogate modelling, optimization under uncertainty, RL, distributed computing, GPU acceleration

Software: Julia, JAX, Python, Pytorch, AMICI, PETab, XGBoost, Optuna, MATLAB, R, Git, Databricks, MLflow, PySpark, SQL, Docker, Azure

PUBLICATIONS

- Pessina, D. et al.** (2026) Transfer Learning of Data-driven Crystallisation Processes via Constrained Neural Ordinary Differential Equations; *Dig. Chem. Eng.*
- Pessina, D. et al.** (2025) Model-based approach to template-induced macromolecule crystallisation; *Systems & Control Transactions*
- Pessina, D. et al.** (2025) Machine learning-enhanced Sensitivity Analysis for Complex Pharmaceutical Systems; *Systems & Control Transactions*
- Pessina, D. et al.** (2025) Integrated In Vitro/In Silico Uncertainty Quantification Method for Protein Crystallization Models; *Industrial & Engineering Chemistry Research*

OPEN-SOURCE CONTRIBUTIONS

- CriSTool** | *Julia*-based simulation platform for crystallisation population balance models with uncertainty quantification
- gsax** | JAX-accelerated package for fast global sensitivity analysis of large-scale dynamic models.